Apricot Reference Laboratories
(800) 555-0151 | www.apricot.example.org
2700 Canyon Way, Ashland, UT 00840
Gregory Lund, MD, PhD, Medical Director

Client: Riverbend Medical Group 230
8525 Harbor Road
Lakewood, NJ 00800
UNITED STATES

Physician: Martin Kowal, MD

Patient: Petrov, Arjun

DOB	2/8/1986
Gender:	Male
Patient Identifiers:	64927575523
Visit Number (FIN):	4658304460
Collection Date:	7/1/2026

COL6A1 c.1611+1G>A Variant Analysis

Apricot test code 9126265

COL6A1 Specimen	Blood

COL6A1 c.1611+1G>A Variant Analysis

Negative

Indication for testing: Diagnostic evaluation for muscle weakness.

NEGATIVE: The *COL6A1* c.1611+1G>A variant was not detected. No reportable variant was found. The test looked only for one known change in the COL6A1 gene, so it does not rule out Bethlem myopathy.

Reviewed and signed out by Sana Qureshi, PhD.

H=High, L=Low, *=Abnormal, C=Critical

Performing laboratory:
Apricot Reference Laboratories
SYNTHETIC TEST REPORT - fictional patient and laboratory - not for clinical use | SYN-0014 | Page 1 of 3

BACKGROUND INFORMATION: COL6A1 c.1611+1G>A Variant Analysis

CHARACTERISTICS: Bethlem myopathy is a muscle disease that causes weakness and joint stiffness. The weakness is usually mild to moderate and often grows slowly. Many people also have contractures, which are joints that do not fully straighten or bend.

INCIDENCE: Bethlem myopathy is rare. Its frequency is not well known, and many cases are probably missed or diagnosed late because the symptoms are mild.

INHERITANCE: Bethlem myopathy is most often inherited in an autosomal dominant pattern, which means one changed copy of a collagen VI gene can cause the condition. Recessive forms also exist. The pattern depends on the gene and the change.

PENETRANCE: Penetrance is high but not complete. Some people who carry a change have few or no symptoms, and the severity can differ between family members who carry the same change.

CAUSE: The condition is caused by changes in the genes that make collagen VI, a protein that forms a support network around muscle fibers. When these changes reduce the amount or quality of collagen VI, the muscle fibers are weaker and more prone to damage.

CLINICAL SENSITIVITY: Depends on the condition and the population tested.

METHODOLOGY: Real-time PCR genotyping of the targeted variant with fluorescent probes. Variants are named according to HGVS nomenclature, for example c.1565A>C for a substitution in the coding sequence.

ANALYTICAL SENSITIVITY AND SPECIFICITY: 99 percent.

LIMITATIONS: In rare cases, an uncommon sequence variant can interfere with the assay and cause an incorrect result. The assay detects only the targeted variant, not other variants in the gene.

This test was developed and its performance characteristics determined by Apricot Reference Laboratories. It has not been cleared or approved by the US Food and Drug Administration.

Ask the laboratory whether a broader analysis of the COL6A1 gene, or of the other collagen VI genes, is available.

Consider a multi-gene panel or exome sequencing if the muscle weakness is still unexplained, because this test checked only one known change.

Review the clinical findings with a neuromuscular specialist, who can decide whether a muscle biopsy or other tests are needed.

VERIFIED/REPORTED DATES

Procedure	Accession	Collected	Received	Verified/Reported
COL6A1 Specimen	57-153-399253	7/1/2026	7/5/2026	8/22/2026
COL6A1 c.1611+1G>A Variant Analysis	57-153-399253	7/1/2026	7/5/2026	8/22/2026

END OF CHART

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Provided Clinical Phenotype

As written by the ordering clinician on the test requisition form.

- Increased endomysial connective tissue
- Elevated blood creatine phosphokinase
- Proximal limb muscle weakness